

Calculus Linear Algebra Practice Test

Name: _____

For the following questions, a calculator may be used.

1. Solve the following system of equations. If there are no solutions, state so and if there are infinite solutions, be sure to describe the relationship of the solutions.

$$\begin{array}{l} x - 3z = -2 \\ \text{a) } 3x + y - 2z = 5 \\ 2x + 2y + z = 4 \end{array}$$

$$\begin{array}{l} x + 2y = 7 \\ \text{b) } 2x + y = 8 \end{array}$$

$$\begin{array}{l} x + y + 4z = 5 \\ \text{c) } 2x + y - z = 9 \end{array}$$

$$\begin{array}{l} 2x + y - z = 2 \\ \text{d) } x + y - 3z = 4 \\ 3x + 2y - 4z = 7 \end{array}$$

2. Find AB and then BA , if possible.

a) $A = \begin{bmatrix} 2 & 1 \\ 4 & -3 \\ 1 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 & 5 \\ 3 & -2 & 0 \end{bmatrix}$

b) $A = \begin{bmatrix} 1 & 7 & -2 \\ 0 & 4 & 3 \end{bmatrix}, B = \begin{bmatrix} -3 \\ 4 \\ 5 \end{bmatrix}$

3. Are the following matrices inverses?

$$N = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \quad , \quad O = \begin{bmatrix} 2 & -17 & 11 \\ -1 & 11 & -7 \\ 0 & 3 & -2 \end{bmatrix}$$

4. Find the inverse of:

a) $\begin{bmatrix} 1 & -2 \\ 2 & -3 \end{bmatrix}$

b) $\begin{bmatrix} 1 & 1 & 1 \\ 3 & 5 & 4 \\ 3 & 6 & 5 \end{bmatrix}$

5. Find the determinant of the following:

a) $\begin{bmatrix} 7 & -2 \\ \frac{1}{2} & -3 \end{bmatrix}$

b) $\begin{bmatrix} 3 \cos \theta & -2 \sin \theta \\ 6 \sin \theta & 4 \cos \theta \end{bmatrix}$

c) $\begin{bmatrix} 1 & 2 & 1 \\ 3 & 5 & -4 \\ 1 & 6 & 5 \end{bmatrix}$

Solutions:

1. a)
$$\begin{aligned} x - 3z &= -2 \\ 3x + y - 2z &= 5 \\ 2x + 2y + z &= 4 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 3 & 1 & -2 & 5 \\ 2 & 2 & 1 & 4 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 7 & 11 \\ 0 & 2 & 7 & 8 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 7 & 11 \\ 0 & 0 & -7 & -14 \end{array} \right] \rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -3 & -2 \\ 0 & 1 & 7 & 11 \\ 0 & 0 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 1 & 2 \end{array} \right] \quad x = 4, y = -3, z = 2$$

b)
$$\begin{aligned} x + 2y &= 7 \\ 2x + y &= 8 \end{aligned}$$

$$\left[\begin{array}{cc|c} 1 & 2 & 7 \\ 2 & 1 & 8 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & -3 & -6 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 7 \\ 0 & 1 & 2 \end{array} \right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & 3 \\ 0 & 1 & 2 \end{array} \right] \quad x = 3, y = 2$$

c)
$$\begin{aligned} x + y + 4z &= 5 \\ 2x + y - z &= 9 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & 4 & 5 \\ 2 & 1 & -1 & 9 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 4 & 5 \\ 0 & -1 & -9 & -1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & 4 & 5 \\ 0 & 1 & 9 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 0 & -5 & 4 \\ 0 & 1 & 9 & 1 \end{array} \right]$$

Which means $\begin{aligned} x - 5z &= 4 \\ y + 9z &= 1 \end{aligned}$, so there are infinite solutions. This can be

expressed parametrically as $x = 4 + 5t, y = 1 - 9t, z = t$

d)
$$\begin{aligned} 2x + y - z &= 2 \\ x + y - 3z &= 4 \\ 3x + 2y - 4z &= 7 \end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 2 & 1 & -1 & 4 \\ 3 & 2 & -4 & 7 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -3 & 2 \\ 0 & -1 & 5 & 0 \\ 0 & -1 & 5 & 1 \end{array} \right] \rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -3 & -2 \\ 0 & 1 & -5 & 0 \\ 0 & -1 & 5 & 1 \end{array} \right] \rightarrow$$

$$\left[\begin{array}{ccc|c} 1 & 1 & -3 & -2 \\ 0 & 1 & -5 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right] \text{ The last line is a contradiction, so no solution.}$$

2.

a)

$$AB = \begin{bmatrix} 2 & 1 \\ 4 & -3 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 & 5 \\ 3 & -2 & 0 \\ 5 & -2 & 10 \\ -5 & 6 & 20 \\ 19 & -12 & 5 \end{bmatrix}$$

$(1)(2) + (3)(1) = 5$
 $(4)(0) + (-3)(-2) = 6$
 $(1)(5) + (6)(0) = 5$

$$BA = \begin{bmatrix} 1 & 0 & 5 \\ 3 & -2 & 0 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 4 & -3 \\ 1 & 6 \\ 7 & 31 \\ -2 & 9 \end{bmatrix}$$

$(1)(2) + (0)(4) + (5)(1) = 7$
 $(3)(1) + (-2)(-3) + (0)(6) = 9$

b)

$$AB = \begin{bmatrix} 1 & 7 & -2 \\ 0 & 4 & 3 \end{bmatrix} \begin{bmatrix} -3 \\ 4 \\ 5 \\ 15 \\ 31 \end{bmatrix}$$

$(1)(-3) + (7)(4) + (-2)(5) = 15$
 $(0)(-3) + (4)(4) + (3)(5) = 31$

$$BA = \begin{bmatrix} -3 \\ 4 \\ 5 \end{bmatrix} \begin{bmatrix} 1 & 7 & -2 \\ 0 & 4 & 3 \\ ? & ? & ? \\ ? & ? & ? \\ ? & ? & ? \end{bmatrix}$$

One number in the row and two in the column, can't multiply.

3. Matrices are inverses if $NO = I_3$

$$NO = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 4 & -3 \\ 3 & 6 & -5 \end{bmatrix} \begin{bmatrix} 2 & -17 & 11 \\ -1 & 11 & -7 \\ 0 & 3 & -2 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{So yes they are inverses}$$

4.

$$\text{a) } \begin{bmatrix} 1 & -2 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 1 & -2 & | & 1 & 0 \\ 2 & -3 & | & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & -2 & | & 1 & 0 \\ 0 & 1 & | & -2 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & | & -3 & 2 \\ 0 & 1 & | & -2 & 1 \end{bmatrix} \quad \begin{bmatrix} -3 & 2 \\ -2 & 1 \end{bmatrix}$$

$$\begin{aligned} \text{b) } & \begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 3 & 5 & 4 & | & 0 & 1 & 0 \\ 3 & 6 & 5 & | & 0 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & 2 & 1 & | & -3 & 1 & 0 \\ 0 & 3 & 2 & | & -3 & 0 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 1 & | & 1 & 0 & 0 \\ 0 & 1 & \frac{1}{2} & | & \frac{-3}{2} & \frac{1}{2} & 0 \\ 0 & 3 & 2 & | & -3 & 0 & 1 \end{bmatrix} \rightarrow \\ & \begin{bmatrix} 1 & 1 & \frac{1}{2} & | & \frac{1}{2} & \frac{0}{2} & 0 \\ 0 & 1 & \frac{1}{2} & | & \frac{-3}{2} & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} & | & \frac{3}{2} & \frac{-3}{2} & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & \frac{1}{2} & | & \frac{1}{2} & \frac{0}{2} & 0 \\ 0 & 1 & \frac{1}{2} & | & \frac{-3}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 & | & 3 & -3 & 2 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 1 & 0 & | & -2 & 3 & -2 \\ 0 & 1 & 0 & | & -3 & 2 & -1 \\ 0 & 0 & 1 & | & 3 & -3 & 2 \end{bmatrix} \rightarrow \\ & \begin{bmatrix} 1 & 1 & 0 & | & 1 & 1 & -1 \\ 0 & 1 & 0 & | & -3 & 2 & -1 \\ 0 & 0 & 1 & | & 3 & -3 & 2 \end{bmatrix} \quad \begin{bmatrix} 1 & 1 & -1 \\ -3 & 2 & -1 \\ 3 & -3 & 2 \end{bmatrix} \end{aligned}$$

5.

$$\text{a) } d = (7)(-3) - (-2)\left(\frac{1}{2}\right) = -20$$

$$\text{b) } d = (3 \cos \theta)(4 \cos \theta) - (-2 \sin \theta)(6 \sin \theta) = 12 \cos^2 \theta + 12 \sin^2 \theta = 12$$

$$\text{c) } d = 1 \begin{vmatrix} 5 & -4 \\ 6 & 5 \end{vmatrix} - 2 \begin{vmatrix} 3 & -4 \\ 1 & 5 \end{vmatrix} + 1 \begin{vmatrix} 3 & 5 \\ 1 & 6 \end{vmatrix} = 1(25 + 24) - 2(15 + 4) + 1(18 - 5) = 24$$